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Market Insurance, Self-Insurance, and Self-Protection

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The article develops a theory of demand for insurance that emphasizes the interaction between market insurance, "self-insurance," and "self-protection." The effects of changes in "prices," income, and other variables on the demand for these alternative forms of insurance are analyzed using the "state preference" approach to behavior under uncertainty. Market insurance and self-insurance are shown to be substitutes, but market insurance and self-protection can be complements. The analysis challenges the notion that "moral hazard" is an inevitable consequence of market insurance, by showing that under certain conditions the latter may lead to a reduction in the probabilities of hazardous events.

The incentive to insure and its behavioral implications have usually been analyzed by applying the expected utility approach without reference to the indifference curve analysis ordinarily employed in consumption theory. In this paper insurance is discussed by combining expected utility and an indifference curve analysis within the context of the "state preference" approach to behavior under uncertainty (the preferences in question relating to states of the world).[1] We use this framework to restate

Becker's contribution was primarily an unpublished paper that sets out the approach developed here. Ehrlich greatly extended and applied that approach and was primarily responsible for writing this paper. We have had many helpful comments from Harold Demsetz, Jacques Drèze, Jack Hirshleifer, and members of the Labor Workshop at Columbia University and the Industrial Organization Workshop at the University of Chicago.

[1] An approach originally devised by Arrow (1963–64) and worked out in application to investment decisions under uncertainty by Hirshleifer (1970).

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and reinterpret in a simpler and more intuitive way some familiar propositions concerning insurance behavior; more important, we derive a number of apparently new results, especially those concerned with self-insurance and self-protection. Our approach separates objective opportunities from "taste" and other environmental factors, which facilitates an independent investigation of each class of factors analytically as well as empirically. In addition, we consider not only the incentive to insure, but also how much insurance is purchased under varying "opportunities"[2] and in view of the existence of the alternatives of self-insurance and self-protection. We use the basic analytical tools employed throughout traditional consumption and production theory.

It has been argued that insurance is different from "ordinary" goods and services because it is not desired per se, but as a means of satisfying more basic needs.[3] Recent developments in consumption theory[4] suggest, however, that the distinction between goods and services purchased in the market and more basic needs they satisfy is not a unique characteristic of insurance, but applies to all goods and services. The demand for the latter is also derived from the needs they satisfy, just as the demand for factors of production in ordinary production theory is derived from their contribution to final products.

The basic needs underlying the purchase of insurance will be identified with consumption opportunities contingent upon the occurrence of various mutually exclusive and jointly exhaustive "states of the world."[5] Market insurance in this approach redistributes income and, consequently, consumption opportunities, toward the less well-endowed states. Self-insurance, however, redistributes income similarly, self-protection has a related effect, and either might be pursued when market insurance was not available. Moreover, optimal decisions about market insurance depend on the availability of these other activities and should be viewed within the context of a more comprehensive "insurance" decision.

[2] Theorems concerning optimal insurance decisions have been derived in two recent contributions by Smith (1968) and Mossin (1968). Our approach differs not only in form but also in substance; for example, in the analysis of the interaction between market insurance, self-insurance, and self-protection.

[3] For example, Arrow (1965) says, "Insurance is not a material good . . . its value to the buyer is clearly different in kind from the satisfaction of consumer's desires for medical treatment or transportation. Indeed, unlike goods and services, transactions involving insurance are an exchange of money for money, not money for something which directly meets needs" (p. 45).

[4] See, for example, Becker and Michael (1970).

[5] By consumption opportunities in each state of the world is meant command over commodities, C_i, produced by combining market goods, X_i, time spent in consumption, t_i, and the "state environment," E_i, via household production functions (for the latter concept see Becker and Michael 1970): $C_{ij} = f_{ij}(X_{ij}, t_{ij}, E_i)$ $j = 1, \ldots, m$ where j refers to different commodities. If the production functions fully incorporate the effects of environment, the utility function of commodities would not depend on which state occurred. In particular, for an aggregate commodity C, $U(C_0) = U(C_1)$ if $C_0 = {}_1$, where 0, 1 denote different states.

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The first part of this paper spells out a model of market insurance and discusses the effects of changes in terms of trade, "income," and other environmental factors on optimal insurance decisions. Self-insurance, self-protection, and a simultaneous determination of the full insurance decision are then discussed in the second and more original part.

I. Market Insurance

We assume for simplicity that an individual is faced with only two states of the world (0, 1) with probabilities p and $1 - p$, respectively, and that his real income endowment in each state is given with certainty by I_0^c and I_1^c, where $I_1^c - I_0^c$ is the prospective loss if state 0 occurs. If income in state 1 can be exchanged for income in state 0 at the fixed rate

$$-\frac{dI_1}{dI_0} = \pi, \tag{1}$$

π can be called the "price of insurance" measured in terms of income in state 1. The amount of insurance purchased in state 0 can be defined as the difference between the actual and endowed incomes:[6]

$$s = I_0 - I_0^c. \tag{2}$$

The expenditure on insurance measured in terms of state 1's income is

$$b = I_1^c - I_1 = s\pi. \tag{3}$$

Substituting (2) in (3) gives the opportunity boundary

$$I_1^c - I_1 = \pi(I_0 - I_0^c), \tag{4}$$

or the line $\overline{AB}$ in figure 1.[7] It is assumed that the individual chooses the optimal income in states 1 and 0 by maximizing the expected utility of the income prospect,

$$U^* = (1 - p)\, U(I_1) + p\, U(I_0),^8 \tag{5}$$

subject to the constraint given by the opportunity boundary. The first-order optimality condition is

$$\pi = \frac{p\, U_0'}{(1 - p) U_1'}, \tag{6}$$

[6] Note that insurance is defined not in terms of the liability "coverage" of potential losses, as in Smith's (1968) and Mossin's (1968) papers, but in terms of "coverage minus premium," or the net addition to income in state 0.

[7] In figure 1, the opportunity boundary $\overline{AB}$ is drawn as a straight line. This assumes that the same terms of trade apply to both insurance and "gambling," that is, to movements to the right and to the left of E, the endowment position. In practice, the opportunity boundary may be kinked about the endowment point.

[8] For analytical simplicity we ignore the time and environment inputs and assume only a single aggregate commodity in each state. Then the output of commodities can be identified with the input of goods and services, or with income.

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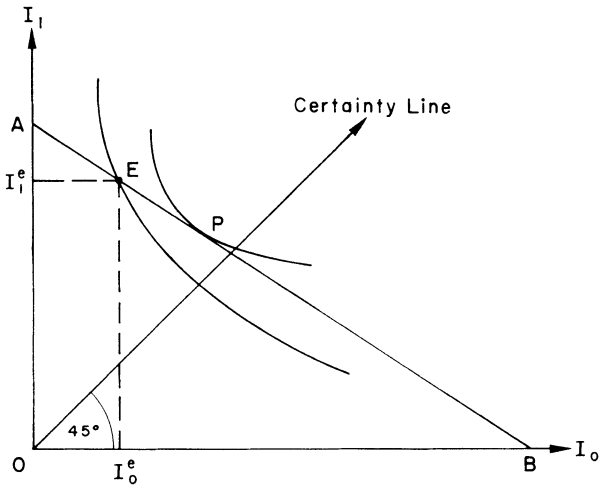


FIG. 1

where $(pU_0')/(1-p)U_1'$ is the slope of the indifference curve (defined along $dU^* = 0$), and π is the slope of the budget line. In equilibrium, they must be the same (see point P).

One can more completely separate tastes from environmental factors by dividing $p/(1-p)$ through in (6) to obtain

$$\bar{\pi} = \frac{1-p}{p}\pi = \frac{U_0'}{U_1'}. \tag{7}$$

Further, $\bar{\pi}$, the price of insurance deflated by the actuarially "fair"[9] price, $p/(1-p)$, is a measure of the "real" price of insurance because a fair price is "costless" to the individual (see the second paragraph below). Equation (7) thus implies that, in equilibrium, the real price of insurance equals the ratio of the marginal utility of I_0 to that of I_1, the ordinary result in consumer demand theory.

The second-order condition requires that the indifference curve be convex to the origin at the equilibrium point, or

$$D = -p\,U_0'' - \pi^2\,(1-p)\,U_1'' > 0. \tag{8}$$

A sufficient condition is that the marginal utility of income is strictly declining.[10]

An immediate implication of equation (7) is that insurance would be

[9] An actuarially fair exchange is an exchange of $p/(1-p)$ units of income in state 1 for an additional unit of income in state 0, where $p/(1-p)$ is the odds that state 0 would occur.

[10] Hirshleifer (1970, p. 233) points out that although diminishing marginal utility of income is not a necessary condition for equilibrum at any given point, it is a necessary condition for the indifference curve to be convex at all points.

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demanded—some I_1 would be traded for I_0—if the slope of the indifference curve exceeded the price of insurance at the endowment point, E:

$$\bar{\pi} < \frac{U'(I_0^e)}{U'(I_1^e)}. \tag{9}$$

If the opposite were true, "gambling" would be demanded, provided similar terms of trade apply in redistributions of income toward state 1. Note that gambling can occur without increasing marginal utility of income if the opportunities available are sufficiently favorable. Therefore, inferences about attitudes toward risk cannot be made independently of existing market opportunities: a person may appear to be a "risk avoider" under one combination of prices and potential losses and a "risk taker" under another.[11]

If the price of insurance were actuarially fair, equation (7) would reduce to $1 = U_0'/U_1'$: incomes would be equalized in both states of the world if the marginal utility of income were always diminishing. This is "full insurance" in the sense that a person would be indifferent as to which state occurred.[12] In particular, for small changes around the equilibrium position, he would act as if he were indifferent toward risk and interested only in maximizing his expected income. Indeed, his income in each state would equal his expected income;[13] therefore, fair insurance can be regarded as costless to him.[14]

[11] Indeed, when faced with several independent hazards, a person might "gamble" and "insure" at the same time, provided the different hazards were associated with different opportunities. For example, given a fair price of theft insurance, he may fully insure his household against theft and at the same time engage in a risky activity if his expected earnings there were greater than his earnings in alternative "safe" activities (see Ehrlich 1970).

[12] Full insurance can be identified with full coverage of potential losses, since the equation $I_1 = I_0$ implies that $I_1^e - b = I_0^e + d - b$, where d is the gross coverage and b is the premium. Clearly, then, $d = I_1^e - I_0^e$. By the same reasoning, since an "unfair" price of insurance $\bar{\pi} > 1$ implies that $I_1 > I_0$, it also implies necessarily less than full coverage of potential losses.

[13] If $I_0 = I_1 = I$ that is, $I_1 = I_1^e - s\pi = I_0 = I_0^e + s$, where $\pi = p/(1-p)$, then $I = p\,I_0^e + (1-p)\,I_1^e$.

[14] Although the model has been developed for two states of the world, the analysis applies equally well to n states. We define the state with the highest income—say, state n—as the state without hazard and define all the states with hazard ($h = 1, \ldots, n-1$) relative to that state. Denoting by p_h the probability of state h, by

$$p = 1 - \sum_{h=1}^{n-1} p_h$$

the probability of state n, and by π_h the implicit terms of trade between income in state n and income in state h, it can easily be shown that if the terms of trade were fair ($\bar{\pi}_h = [p/p_h]\ \pi_h = 1$) s_h would be chosen to equalize incomes in all states of the world and losses would be "fully covered." If the real terms of trade were unfair but

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A. *Substitution Effects*

The effect of an exogenous increase in the price of insurance on the demand for I_0, with the probability of loss and the initial endowment being the same, can be found by partially differentiating the first-order optimality condition with respect to π:

$$\frac{\partial I_0}{\partial \pi} = \frac{1}{D}\left[-(1-p)\,U_1' + (I_0 - I_0^e)\,\pi\,(1-p)\,U_1''\right]. \quad (10)$$

Since the denominator D has already been shown to be positive, the sign of equation (10) is the same as the sign of the numerator, or negative if $I_0 > I_0^e$, since we are assuming $U_1'' < 0$. An increase in the relative cost of income in state 0 necessarily decreases the demand for income in this state. Moreover, it also reduces the amount of insurance purchased, since I_0^e remains unchanged: $\partial s/\partial \pi = \partial I_0/\partial \pi - \partial I_0^e/\partial \pi = \partial I_0/\partial \pi$.

Similarly, the effect of an increase in π on I_1, and thus on the amount spent on insurance, is

$$\frac{\partial I_1}{\partial \pi} = \frac{1}{D}\left[(1-p)\,U_1'\,\pi + (I_0 - I_0^e)\,p\,U_0''\right]. \quad (11)$$

Here the result is ambiguous since U_1' is positive whereas

$$(I_0 - I_0^e)\,pU_0'' = spU_0''$$

is negative if $U_0'' < 0$ and $s > 0$. The result is ambiguous because, although an increase in π reduces the amount of insurance purchased, each unit purchased becomes more expensive. Consequently, the amount spent on insurance would decline only if the price elasticity of demand for insurance exceeded unity[15] (a proof is obvious).

Equations (10) and (11) do not isolate a "pure" substitution effect because an increase in π lowers the opportunities available (if $s > 0$). If both I_1 and I_0 are superior goods, the income and substitution effects both reduce the demand for I_0, whereas they have opposite effects on the demand for I_1. Diagrammatically, as the opportunity boundary changes from $\overline{AB}$ to $\overline{CD}$ (see fig. 2), the equilibrium point shifts from P to Q. If I_0 were a superior good, Q must be to the left of P'. Even if I_0 were an inferior good, however, a "pure" (that is, expenditure-compensated) increase in the terms of trade must always reduce the demand for I_0 and increase I_1: the equilibrium must shift from P to a point to its left, like S.

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constant ($\bar{\pi}_h = \pi = 1 + \lambda > 1$ for all h), s_h would be chosen to equalize incomes in states with hazard only, that is, we would achieve what has been called full insurance above a deductible (for a definition of this concept and an alternative proof see Arrow 1963).

[15] This analysis, therefore, also shows that the effect of a change in π on the "fullness" of insurance (the difference $I_1 - I_0$) and thus on the degree of gross coverage is generally not unambiguous.

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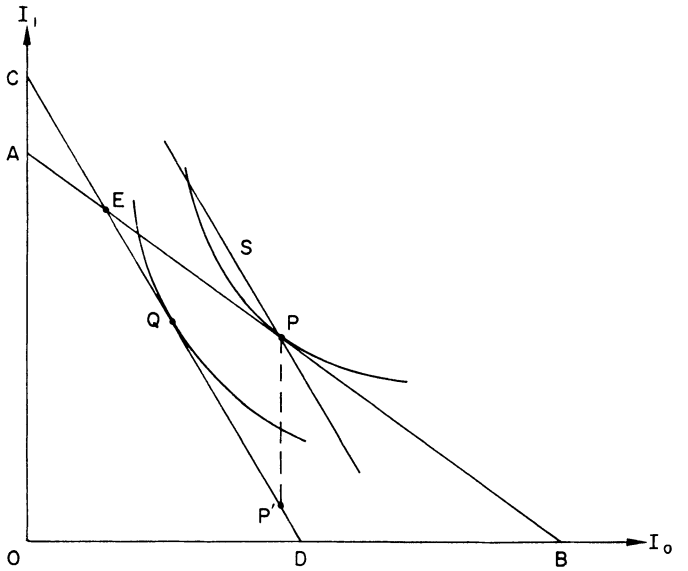


FIG. 2

B. Income Effects

Equation (4) can be written as

$$I_1^e + \pi I_0^e = W = I_1 + \pi I_0, \tag{12}$$

where W is a measure of the total opportunities available. (This is shown in fig. 1 by the intercept $\overline{OA}$ on the I_1 axis.) The effect of a change in the endowments on the income demanded in each state can be determined by differentiating the first-order condition:

$$\begin{aligned} \frac{\partial I_0}{\partial W} &= \frac{\partial I_0}{\partial I_0^e} = \frac{\partial I_0}{\partial I_1^e}\frac{\partial I_0^e}{\partial W} = -\frac{D_{31}}{D} \\ \frac{\partial I_1}{\partial W} &= \frac{\partial I_1}{\partial I_0^e} = \frac{\partial I_1}{\partial I_1^e}\frac{\partial I_0^e}{\partial W} = -\frac{D_{32}}{D} \end{aligned} \tag{13}$$

where $D_{31} = \pi(1 - p)U_1''$, and $D_{32} = pU_0''$. The income demanded in each state necessarily increases with opportunities if the marginal utility of income is falling. Hence, an increase in each state's endowment increases the demand for income in other states as well. The effects on the demand for insurance are more complicated, however, since they depend on how different endowments change. For example, if I_1^e alone increased,

$$\frac{\partial s}{\partial I_1^e} = \frac{\partial I_0}{\partial I_1^e} > 0, \tag{14}$$

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and the demand for insurance would increase. Similarly, if I_0^e alone increased,

$$\frac{\partial s}{\partial I_0^e} = \frac{\partial I_0}{\partial I_0^e} - 1 < 0,^{16} \tag{15}$$

and the demand for insurance would decrease. Equations (14) and (15) imply that if the difference in endowed income—the endowed loss from the hazard—increased either because I_0^e decreased or I_1^e increased, the demand for insurance would increase. Put differently, a person would be more likely to insure large rather than small losses (see Lees and Rice 1965).[17] The effects of a change in total opportunities on the demand for insurance cannot be derived without knowledge of the way opportunities change essentially because insurance is a "residual" that bridges the gap between endowed and desired levels of income in different states of the world.[18]

For example, if both endowments (and hence the size of the loss) are changed by the same percentage, then

$$(\varepsilon_{sW} - 1) = \frac{I_0}{s}(\eta_0 - 1), \tag{16}$$

where $\varepsilon_{sW} = \partial s/\partial W \cdot W/s$ and $\eta_0 = \partial I_0/\partial W \cdot W/I_0$ are the opportunity elasticities of demand for s and I_0, respectively.[19]

Equation (16) incorporates the rather obvious conclusion that the effect of a change in opportunities on the demand for insurance depends on the effects on the income demanded in each state. If the slopes of the indifference curves are constant along a given ray from the origin (the indifference curves are like EPF and GQ_1H in fig. 3)—there is constant

[16] According to equation (13), $\partial I_0/\partial I_0^e = [-\pi^2(1-p)U_1'']/[-pU_0'' - \pi^2(1-p)U_1''] = \delta$, where clearly $0 < \delta < 1$ if U_1'' and $U_0'' < 0$. But since $s = I_0 - I_0^e$, $\partial s/\partial I_0^e = \delta - 1 < 0$.

[17] Similarly, he would be less likely to take large gambles (see the discussion in Hirshleifer 1966). Of course, if insurance is fair he will fully insure all losses, large or small.

[18] Note the analogy between insurance and savings: the latter bridges the gap between "endowed" and desired levels of consumption at different points in time.

[19] Given $s = I_0 - I_0^e$ and $I_1^e = \gamma I_0^e$, then $(\partial s/\partial W)(W/s) = (ds/dI_0^e)(I_0^e/s) = \eta_0 (I_0/s) - (I_0^e/s)$; by collecting terms, we get equation (16). Since $I_0 \gtrless s, e_{sW} \gtrless 1$ if $\eta_0 \gtrless 1$.

If the loss is unaffected by an equal increase in endowments, that is, if $I_0^e = I_1^e - L$ where L is a constant, then $\varepsilon_{se} = (ds/dI_1^e)(I_1^e/s) = (I_0/s)\eta_0 (d \log W)/(d \log I_1^e) - (I_1^e/s)$. This implies that

$$\varepsilon_{se} \begin{matrix} > \\ = \\ < \end{matrix} 0 \quad \text{as} \quad \eta_0 \begin{matrix} > \\ = \\ < \end{matrix} \frac{W}{I_0(1+\pi)} \gtrless 1 \quad (\text{if} \quad I_1 \gtrless I_0). \tag{16a}$$

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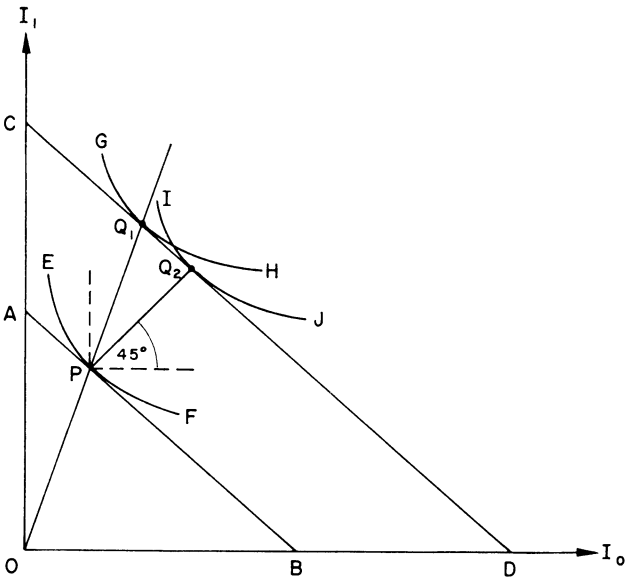


FIG. 3

relative risk aversion[20]—then all equilibrium positions lie on a given ray from the origin, as P and Q_1 do in figure 3, and $\eta_0 = \eta_1 = 1$. An equal proportional increase in all endowments would then increase the demand for insurance by the same proportion. If the slopes of the indifference

[20] Note that

$$\frac{d \text{ slope}}{dI_0} = \frac{d}{dI_0}\left[\frac{pU_0'}{(1-p)U_1'}\right]$$

subject to $I_1 = \gamma I_0$ $is \gtreqless 0$ as $-(U_1''/U_1') \gtreqless -(U_0''/U_0')$; the latter defines increasing, constant, cr decreasing relative risk aversion. Similarly

$$\frac{d}{dI_0}\left[\frac{pU_0'}{(1-p)U_1'}\right]$$

subject to $I_1 - I_0 = L$ is $\gtreqless 0$ as $-(U_1''/U_1') \gtreqless -(U_0''/U_0')$; the latter defines increasing, constant, or decreasing absolute risk aversion (see Pratt 1964, Arrow 1965). (Diagramatically, constant absolute risk aversion implies that the slopes of the indifference curves are constant along any 45° line joining two equilibrium positions—the indifference curves are like EPF and IQ_2J in fig. 3.) Equation (16a) in n.19 implies that increasing relative risk aversion, $\eta_0 > 1$, is compatible with decreasing absolute risk aversion, $\epsilon_{se} < 0$, only if $\eta_0 < W/I_0(1+\pi)$.

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curves increase along a given ray from the origin, as shown by *EPF* and IQ_2J in figure 3, there is increasing relative risk aversion, and η_0 and thus ε_{sW} would exceed unity. Increasing relative risk aversion implies that the elasticity of substitution between I_0 and I_1 tends to decline as opportunities increase.[21] Regardless of the shape of preferences elsewhere in the preference space, however, relative (and absolute) risk aversion remains constant along the certainty line. This constancy always characterizes choices when the price of insurance is actuarially fair (see fig. 4).

C. *Rare Losses*

An inspection of the necessary conditions for insurance given in equation (9) shows that changes in p, the probability of loss, do not affect the incentive to insure as long as the real price of insurance is independent of p. If insurance were actuarially fair, the real price would always equal unity, and thus would be independent of p.

The deviation from a fair price, or the "loading" in insurance terminology, can be defined from the identity

$$\pi \equiv \frac{(1+\lambda)\,p}{1-p}, \tag{17}$$

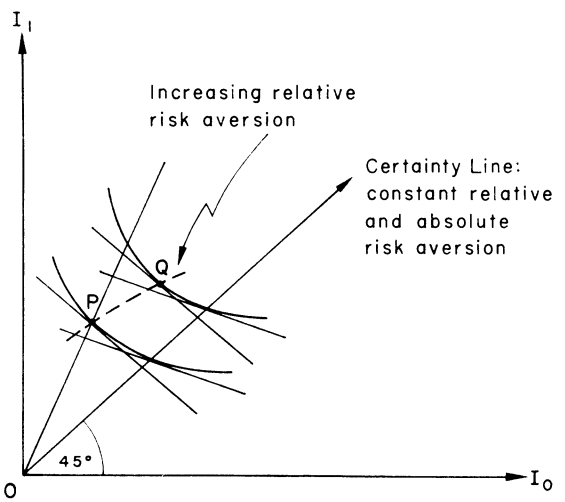


FIG. 4

[21] Since the slopes of the indifference curves necessarily are constant along the "certainty line" and by assumption become increasingly steep toward I_0 along other rays from the origin, a given percentage deviation of the price of insurance from the fair price results in smaller percentage changes in the ratio I_1/I_0 at higher indifference levels. That is, $\sigma = (d \log I_1/I_0)/d \log \pi$ decreases at higher indifference levels when π equals the fair price.

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where λ is the "loading factor." If λ were independent of p, so also would be the real price of insurance and p would have no effect on the incentive to insure. In particular, there would *not* then be a greater incentive to insure "rare" losses of a given size.[22]

Since, apparently, rare losses are more frequently insured,[23] λ is presumably positively related to p, perhaps because processing and investigating costs increase as p increases.[24] (An alternative explanation is provided by the interaction between market and self-insurance analyzed in the next section.) Even if the *incentive* to insure were independent of p, the *amount* insured would decline and the expenditure on insurance would increase as p increased.[25]

II. Self-Insurance and Self-Protection

Two alternatives to market insurance that have not been systematically analyzed in the literature on insurance are self-insurance—a reduction in the size of a loss—and self-protection—a reduction in the probability of a loss.[26] For example, sprinkler systems reduce the loss from fires; burglar alarms reduce the probability of illegal entry; cash balances reduce fluctuations in consumption; medicines, certain foods, and medical

[22] This result appears to contradict one by Lees and Rice (1965) because they define the loading factor in terms of the gross rather than net amount paid in claim; that is by λ' in $\pi = [(1+\lambda')\,p]/[1-(1+\lambda')\,p]$. A reduction in p, λ' held constant, would reduce λ—our definition of the loading factor—and thus would increase the incentive to insure.

[23] Some evidence is presented in Lees and Rice (1965).

[24] Let the amount a that is spent processing and investigating each claim be the only administrative cost of providing insurance. In a zero profit equilibrium position, the unit price of insurance would equal the ratio of the total amount collected in premiums in state 1 (including administration costs) to the difference between the net amount paid in claims in state 0 and administration costs: $\pi = (p\,d + p\,a)/[d\,(1-p) - p\,a]$, where d is the amount covered by insurance. The degree of loading defined by $\lambda = [(1-p)/p]\,\pi - 1 = (d+a)/[d - p\,a/(1-p)] - 1 = a/[d - p\,(d+a)]$ would be larger the larger p was if d were fixed (d would tend to decrease as p increased, and this would increase λ even further).

[25] Generally, the effect of an increase in p on the optimal values of I_0 and I_1, assuming that $\pi = [(1+\lambda)\,p]/(1-p)$ and that λ, I_0^e, and I_1^e are constant, is given by

$$\frac{\partial I_0}{\partial p} = \frac{1}{D}\left[U_1''\,s\,\pi\,\frac{1+\lambda}{1-p}\right] < 0$$

$$\frac{\partial I_1}{\partial p} = \frac{1}{D}\left[U_0''\,s\,\pi\,\frac{1}{1-p}\right] < 0$$

provided $U'' < 0$. An increase in p would then lower the optimal amount of insurance $s = I_0 - I_0^e$ and increase the optimal expenditure on insurance $b = I_1^e - I_1$.

[26] These have been called "loss protection" and "loss prevention," respectively (see Mehr and Commack 1966, pp. 28–29).

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checkups reduce vulnerability to illness; and good lawyers reduce both the probability of conviction and the punishment for crime. As these examples indicate, it is somewhat artificial to distinguish behavior that reduces the probability of a loss from behavior that reduces the size of a loss, since many actions do both. Nevertheless, we do so for expository convenience and because self-insurance clearly illustrates the insurance principle of redistributing income toward less favorable states.

A. Self-Insurance

Assume that market insurance is unavailable and write the loss to a person as $L = L(L^e, c)$, where $L^e = I_1{}^e - I_0{}^e$ is the endowed loss, c is the expenditure on self-insurance, and $\dfrac{\partial L}{\partial c} = L'(c) \leqslant 0$. The expected utility can be written as

$$U^* = (1 - p)\, U(I_1{}^e - c) + p\, U(I_1{}^e - L(L^e, c) - c).^{27} \qquad (18)$$

The value of c that maximizes equation (18), c^0, satisfies the first-order condition

$$-\frac{1}{L'(c^0) + 1} = \frac{p\, U_0'}{(1 - p)\, U_1'}. \qquad (19)$$

This maximizes expected utility if the marginal utility of income and the marginal productivity of self-insurance are decreasing, that is, if the indifference curves are convex and if the production transformation curve between income in states 1 and 0 (TN in fig. 5) is concave to the origin.[28] A necessary condition for a positive amount of self-insurance obviously is $-L'(c^0) > 1$, or that there be a net addition to income in state 0. A sufficient condition, if the transformation and indifference curves do not have kinks, is that

$$-\frac{1}{L'(L^e, 0) + 1} < \frac{p\, U'(I_0{}^e)}{(1 - p)\, U'(I_1{}^e)}.^{29} \qquad (20)$$

An increase in the unit cost of self-insurance, measured by the marginal

[27] For analytical convenience we assume that $I_0{}^e$ alone is affected by c, although, of course, both endowments may be affected. Moreover, the assumption that $\partial L/\partial c \leqslant 0$ is not always true: an individual could increase $I_1{}^e$ and reduce $I_0{}^e$ by deliberately exposing himself to hazards; for example, by committing a crime or engaging in a risky legal occupation (see Ehrlich 1970). The condition $\partial L/\partial c > 0$ can be said to define "negative self-insurance."

[28] See equation (A5). Note that the transformation curve may be kinked at the endowment point.

[29] If the opposite were true, there would be an incentive to increase the loss by increasing I_1 and reducing I_0 (see n. 27 above).

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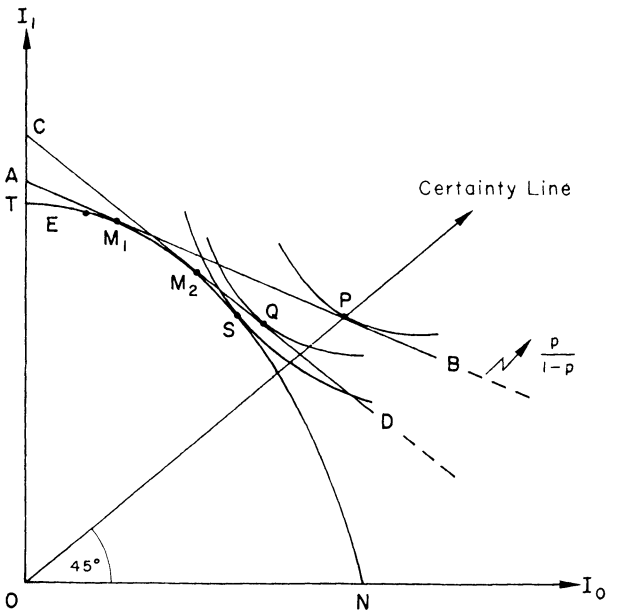


FIG. 5

productivity of self-insurance, would reduce the demand for self-insurance, measured by c^0:[30]

$$\frac{\partial c^0}{\partial \alpha} < 0,^{31} \tag{21}$$

where α is a parameter that reduces the absolute value of L' for a given c. Similarly, a reduction in $I_0{}^e$ would increase the demand for self-insurance:

$$-\frac{\partial c^0}{\partial I_0{}^e} > 0.^{32} \tag{22}$$

Equation (20) shows clearly that the incentive to self-insure, unlike the incentive to use market insurance, is smaller for rare losses. The reason is that the loading factor of self-insurance is *larger* for rare losses because

[30] Although c denotes the expenditure on self-insurance rather than the reduction in the size of the loss, there is a one-to-one relationship between expenditure and insurance because $-L'(c) > 1$.

[31] By differentiating equation (19) with respect to α—p, $I_0{}^e$ and $I_1{}^e$ held constant—one obtains $\partial c^0/\partial \alpha = (pU_0{}'/U^*_{cc})(\partial L'/\partial \alpha) = (+)/(-) < 0$, where $U^*_{cc} = \partial^2 U^*/\partial c^2 < 0$ (see Appendix A), and by assumption $\partial L'/\partial \alpha > 0$.

[32] By differentiating equation (19) with respect to $I_0{}^e - I_1{}^e$, p and L' held constant, one obtains $-\partial c^0/\partial I_0{}^e = \partial c^0/\partial L^e = -\{pU_0{}''[L'(c) + 1]/U^*_{cc}\}\ \partial L/\partial L^e = (-)/(-) > 0$, where by assumption $\partial L/\partial L^e > 0$.

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its price, unlike the price of market insurance, can be presumed to be independent of the probability of loss.[33]

An increase in endowed incomes that resulted from investment in human capital would probably be associated with an increase in the marginal productivity of self-insurance.[34] Therefore, the effect on self-insurance of a change in income has to be separated from the effect of the associated change in marginal productivity.

If market and self-insurance were both available, values of c and s would be chosen simultaneously to maximize the expected utility function,

$$U^* = (1 - p)\, U(I_1^e - c - s\,\pi) + p\, U(I_1^e - L(L^e, c) - c + s). \tag{23}$$

If the price of market insurance were independent of the amount of self-insurance, the first-order optimality conditions would be

$$\begin{aligned} &-(1 - p)\, U_1'\,\pi + p\, U_0' = 0. \\ &-(1 - p)\, U_1' - p\, U_0'[L'(c) + 1] = 0. \end{aligned} \tag{24}$$

By combining these equations we get

$$\pi = -\frac{1}{L'(c) + 1}. \tag{25}$$

In equilibrium, therefore, the "shadow price" of self-insurance would equal the price of market insurance.

Clearly, market insurance and self-insurance are "substitutes" in the sense that an increase in π, the probability of loss being the same, would decrease the demand for market insurance and increase the demand for self-insurance.[35] For example, a change in the market insurance line from $\overline{AB}$ to $\overline{CD}$ in figure 5 would increase self-insurance by the horizontal distance between M_1 and M_2 and reduce market insurance by the horizontal distance between Q and P. In particular, the purchase of market insurance would reduce the demand for self-insurance—compare points S and, say, M_1.

When market insurance is available at a fair price, the equilibrium condition (25) becomes

$$-\frac{1}{L'(c) + 1} = \frac{p}{1 - p}, \tag{26}$$

or

[33] The price of self-insurance is given by $\pi = -1/[L'(c) + 1]$, where $L'(c)$ presumably does not depend on p. The loading factor is then given by $\lambda = -\{1/[L'(c) + 1]\}\ [(1 - p)/p] - 1$. Hence $\partial\lambda/\partial p < 0$.

[34] That is, not only would $\partial I_i^e/\partial E > 0\ i = 0, 1$ where E is the stock of human capital, but probably also $\partial^2 L/\partial c \partial E < 0$.

[35] A mathematical proof can be found in Appendix A.

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$$-L'(c) = \frac{1}{p},$$

precisely the condition that maximizes expected income.[36] Even with diminishing marginal utility of income, a person would act as if he were risk neutral and choose the amount of self-insurance that maximized his expected income. Consequently, apparent attitudes toward risk are dependent on market opportunities, and real attitudes cannot easily be inferred from behavior.

More generally, even if the price of market insurance were not fair, the optimal amount of self-insurance would maximize the market value of income (given by W in equation [12]), and would not depend on the shape of the indifference curves or even on the probability distribution of states.[37] Geometrically, optimal self-insurance is determined by moving along the transformation curve in figure 5 to the point of tangency between this curve and a market insurance line; since the market value of income is the intercept on the y-axis, that intercept would be maximized at such point of tangency.

The effects of specific parameters on the demand for market and self-insurance when both are available often are quite different from their effect when market insurance or self-insurance alone is available. For example, although an increase in the endowed loss increases the demand for self- or market insurance when either alone is available since an increase in market insurance itself reduces self-insurance, and vice versa, the indirect effects can offset the direct effects when both market and self-insurance are positive (see Appendix A for an example of this). Similarly, because a decrease in the probability of loss with no change in the market loading factor reduces the demand for self-insurance, it increases the demand for market insurance. Therefore, people may be more likely to use the market to insure rare losses not necessarily because of a positive relation between the probability of loss and the loading factor (see the discussion in Section IC), but because of a substitution between market and self-insurance.

B. Self-Protection, Subjective Probabilities, and "Moral Hazard"

Self-insurance and market insurance both redistribute income toward hazardous states, whereas self-protection reduces the probabilities of these states. Unlike insurance, self-protection does not redistribute income, because the amount spent reducing the probability of a loss decreases income

[36] Equation (26) can be derived by maximizing $(1 - p)\ (I_1{}^c - c) + p\ [I_1{}^c - L(L^c, c) - c]$ with respect to c.

[37] Equation (25) can be derived by maximizing $W = (I_1{}^c - c) + \pi\ [I_1{}^c - L(L^c, c) - c]$ with respect to c. We are indebted to Jacques Drèze for emphasizing this point.

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in all states equally, leaving unchanged the absolute size of the loss (its relative size actually increases).

Studies using the states-of-the-world approach to analyze decision making under uncertainty have assumed that the probability of a state is entirely determined by "nature" and is independent of human actions. With this approach there is no such thing as self-protection; the activities we call by this name would be subsumed under self-insurance. It has been claimed that states can always be defined to guarantee the independence of their probabilities from human actions,[38] but we deny that this can be done in a meaningful way. Consider, for example, the probability that a given house will be damaged by lightning.[39] Since this probability can be reduced by the installation of lightning rods, independent state probabilities could be obtained only by using a more fundamental state description: the probability of a stroke of lightning itself. If control of the weather is ruled out, the probability of lightning can be assumed to be unaffected by human actions. We are concerned, however, about the probability of damage to the house—we do not care about the probability of lightning per se—and the probability of damage is affected by lightning rods.

In other words, although an appropriate definition of states would produce state probabilities that are independent of human actions, it would not produce a probability distribution of outcomes—the relevant probability distribution—that is independent of these actions. Since one of the main purposes of the state-of-the-world approach is to equate the probability distribution of outcomes with the probability distribution of states, a search for state probabilities that are independent of human actions would be self-defeating.

To look at the difference between self-protection and self-insurance from the viewpoint of outcomes, assume the probability distribution of endowed outcomes given by AB in figure 6. Self-insurance, by contracting the distribution to, say, CD, lowers the probability of both high and low outcomes, thereby unambiguously reducing the dispersion of outcomes. Self-protection, on the other hand, by shifting the whole distribution to the left to, say, EF, reduces the probability of low outcomes and raises the probability of high ones and does not have an unambiguous effect on the dispersion.[40]

Since the preceding discussion shows that self-insurance is to be distinguished from self-protection, we develop a formal analysis of the latter. Let us assume that the probability of a hazardous state can be reduced by

[38] The only explicit discussion is by Hirshleifer (1970, p. 217).

[39] This example is discussed by Hirshleifer (1970).

[40] The effect of the introduction of self-protection on the variance of income, $p(1-p)\ (I_1^c - I_0^c)^2$, can be found by differentiation $v'(r) \equiv \partial \operatorname{Var}(I^c)/\partial r = (1 -$

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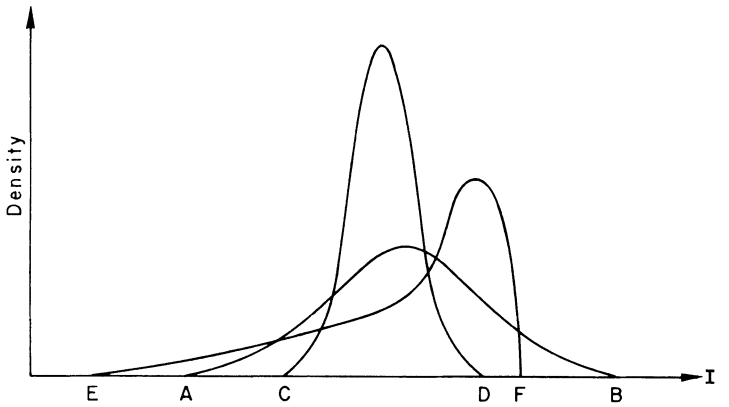


FIG. 6

appropriate expenditure: $p = p(p^e, r)$, where p^e is the endowed probability of hazard, r is the expenditure on self-protection, and $\partial p/\partial r = p'(r) \leqslant 0$. If no market or self-insurance were available, the optimal expenditure on self-protection would maximize

$$U^* = [1 - p(p^e, r)]\, U(I_1{}^e - r) + p(p^e, r)\, U(I_0{}^e - r); \qquad (27)$$

the optimality condition is

$$-p'(r^0)\,(U_1 - U_0) = (1 - p)\,U_1' + p\,U_0'. \qquad (28)$$

The term on the left is the marginal gain from the reduction in p; that on the right, the decline in utility due to the decline in both incomes, is the marginal cost. In equilibrium, of course, they must be equal.

The second-order optimality condition requires that

$$U^*_{rr} = -p''(r^0)\,(U_1 - U_0) + 2p'(r^0)(U_1' - U_0') + (1 - p)\,U_1'' + p\,U_0'' < 0. \qquad (29)$$

Decreasing marginal utility of income is neither a necessary nor a sufficient condition. If $p''(r^0) > 0$, equation (29) is always satisfied if the marginal utility of income is constant and may or may not be satisfied if the marginal utility is decreasing or increasing. This shows that the incentive to self-protect, unlike the incentive to insure, is not so dependent

---

$2p)\ (I_1{}^e - I_0{}^e)^2\, p'(0)$, where r is the expenditure on self-protection. Clearly $v'(r) \gtreqless 0$ as $p \gtreqless 1/2$.

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on attitudes toward risk, and could be as strong for risk preferrers as for risk avoiders.

As with market and self-insurance, the effect of a change in incomes on the demand for self-protection depends on the source of the change as well as on preferences.[41] A decline in I_0^e alone might not increase the demand for self-protection, even if the marginal utility of income were falling, because a decline in I_0^c would increase the marginal cost of self-protection.[42]

A decline in the marginal productivity of self-protection—an increase in the shadow price of protection—always decreases the demand for self-protection regardless of attitudes toward risk.[43] Therefore, if the endowed probabilities and incomes were the same, more efficient providers of self-protection would have lower equilibrium probabilities of hazard. Consequently, different persons use different probabilities in their decision-making process not only because of differences in "temperament," or optimism, but also because of differences in productivity at self-protection. As suggested in the last section, differences in productivity, in turn, may be attributed to differences in education and other forms of "human capital."

If market insurance and self-protection were jointly available, the function

$$U^* = [1 - p(p^e, r)]\, U(I_1^e - r - s\,\pi(r)) + p(p^e, r)\, U(I_0^c - r + s) \tag{30}$$

would be maximized with respect to r and s; the first-order optimality conditions are

$$-(1-p)\, U_1'\,\pi + p\, U_0' = 0 \tag{31}$$

$$-p'(r^*)(U_1 - U_0) - (1-p)\, U_1'[1 + s^*\,\pi'(r^*)] - p\, U_0' = 0. \tag{32}$$

The term $\pi'(r^*)$ measures the effect of a change in self-protection on the price of market insurance through its effects on p and the loading factor λ. From the definition of π in equation (17) we obtain:

$$\frac{\partial \pi}{\partial r} = \pi'(r) = \pi'(p)\, p'(r) + \pi'(\lambda)\, \lambda'(r). \tag{33}$$

[41] An equal proportional increase in endowments ($I_1^e = \gamma I_0^e$) would increase the demand for self-protection if $(dr^0/dI_0^c) = (1/U^*_{rr})[p'(r^0)(U_1'\gamma - U_0') + (1-p)\, U_1''\,\gamma + p\, U_0''] > 0$. A sufficient condition if $U'' < 0$ is $(U_1'/U_0')\ (I_1^c/I_0^e) \geqslant 1$, or that the "average relative risk aversion" between I_0^e and I_1^c be sufficiently greater than one.

[42] That is, $-\partial r^0/\partial I_0^c = [p'(r^0)\, U_0' - pU_0'']/U^*_{rr} \gtreqless 0$ as $-[p'(r^0)]/p \gtreqless -U_0''/U_0'$.

[43] That is, $\partial r^0/\partial\beta = [(U_1 - U_0)/U^*_{rr}]\ (\partial p'/\partial\beta) = (+)/(-) > 0$, where by assumption $\partial p'/\partial\beta < 0$.

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The effect of market insurance on the demand for self-protection has generally been called "moral hazard." In particular, moral hazard refers to an alleged deterrent effect of market insurance on self-protection[44] that increases the actual probabilities of hazardous events (Arrow 1962). Consequently, moral hazard is said to be "a relevant cost of producing insurance that is imposed by the insured on the insurance company" (Demsetz 1969, p. 7) and to provide a "limit to the possibilities of insurance" (Arrow 1962, p. 612). We showed in the last section that market insurance does reduce self-insurance, but no one has shown rigorously why, or under what conditions, market insurance reduces self-protection.

Market insurance has two opposite effects on self-protection. On the one hand, self-protection is discouraged because its marginal gain is reduced by the reduction of the difference between the incomes and thus the utilities in different states (see equation [28]); on the other hand, it is encouraged if the price of market insurance is negatively related to the amount spent on protection through the effect of these expenditures on the probabilities. Consider the relative importance of these opposite effects in two extreme cases:

If market insurance were always available at an actuarially fair price regardless of the amount spent on self-protection, then $\pi = p/(1 - p)$, and equation (31) implies that the optimal amount of market insurance (s^*) equalizes income in both states of the world. There is still an incentive to spend on self-protection, however, because π is negatively related to these expenditures (r):

$$\pi'(r) = \frac{p'(r)}{(1 - p)^2}. \tag{33a}$$

Substituting $U_1 = U_0$ and $U_1' = U_0$ into equation (32), and using equation (33a) and the fact that $s^* = (1 - p)\ (I_1^c - I_0^c)$, we get

$$p'(r^*) = -\frac{1 - p}{s^*} = -\frac{1}{I_1^c - I_0^c}, \tag{34}$$

precisely the condition to maximize expected income. As with self-insurance, a fair price of market insurance encourages an expenditure on self-protection that maximizes expected income. Consequently, moral hazard would not then increase the real cost of insurance, reduce an economy's technical efficiency, or limit the development of market insurance since an

[44] See, for example, Arrow (1962, pp. 612, 613, 616; 1963, pp. 945, 961). Some writers have viewed moral hazard, in part, as a moral phenomenon related to fraud in the collection of benefits (see, for example, Mehr and Commack 1966, p. 174): a fire insurance policy, for example, may create an incentive for arson as well as for carelessness. Our analysis deals explicitly only with the effects of market insurance on self-protection, although implicitly it applies also to the effects on fraud.

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amount of market insurance equalizing income in all states would be chosen.

Even more important is that, contrary to the moral hazard argument, the optimal expenditure on self-protection, r^*, can be *larger* than the amount spent in the absence of market insurance, r^0. By equations (34) and (28), and the condition $p'(r) < 0$, r^* would be larger than r^0 if

$$\frac{U(I_1{}^e - r^0) - U(I_0{}^e - r^0)}{I_1{}^e - I_0{}^e} = \bar{U}' < (1 - p)\, U'(I_1{}^e - r^0) + p\, U'(I_0{}^e - r^0), \tag{35}$$

which is likely provided p is not very small and U is concave. Indeed, if utility were a quadratic function of income, r^* would be larger than r^0 if p were larger than one-half.[45] Not only are market insurance and self-protection complements in the sense that the availability of the former could increase the demand for the latter, but also in the sense that an increase in the productivity of self-protection or a decrease in the real cost of market insurance would increase the demand for both (see Appendix B).

Suppose, at the other extreme, that the price of market insurance was independent of expenditures on self-protection—the loading factor increased sufficiently to offset exactly the reduction in the probability of loss. Self-protection would then usually be discouraged by market insurance—moral hazard would exist—because the main effect of introducing market insurance would be to narrow the differences between incomes in different states.[46] Moreover, since the demand for market insurance is negatively related to the degree of loading, it would be negatively related to expenditures on self-protection. Consequently, for those kinds of market insurance with prices that are largely independent of expenditures on self-protection, one should observe either a large demand for insurance and a small demand for self-protection, or the converse. In our judgment, this

[45] If $U = a\,I + b\,I^2$, with $b < 0$, equation (35) becomes $[p - (1/2)]\, I_1{}^e - [p - (1/2)]\, I_0{}^e > 0$. Since $I_1{}^e > I_0{}^e$, this implies that $p > 1/2$.

[46] If $\pi'(r) = 0$, the optimality condition for r, given the value of s, is from equation (32): $-p'(r^*)\,[U(I_1{}^e - r^* - s\,\pi) - U(I_0{}^e - r^* + s)] - (1 - p)\, U'(I_1{}^e - r^* - s\,\pi) - p\, U'(I_0{}^e - r^* + s) = 0$. Self-protection would be discouraged by market insurance if an exogenous increase in the latter always reduced the optimal value of r^*; that is, if $dr^*/ds < 0$, or $dr^*/ds = \{p'(r^*)\,[U_1{}'(-\pi) - U_0{}'] + [(1 - p)\, U_1{}''(-\pi) + pU_0{}'']\}/U^*{}_{rr} < 0$, where $U^*{}_{rr} < 0$. The first term in the numerator is necessarily positive since $p'(r^*)$ is negative and π, $U_0{}'$ and $U_1{}'$ are all positive. Therefore, a sufficient condition for the inequality to hold is that the second term be nonnegative, or since $\pi = [(1 + \lambda)\, p]/(1 - p)$, that $U_0{}'' \geqslant (1 + \lambda)\, U_1{}''$. If $\lambda \geqslant 0$—no negative loading—this latter inequality necessarily holds provided U'' and U''' $\leqslant 0$; for example, if U were the quadratic function $a\,I + b\,I^2$, with $b \leqslant 0$. Of course, it *might* hold even if $U''' > 0$.

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is the major reason why certain kinds of hazards, like failure in business, are not considered insurable by the market.

Since the price of self-insurance is independent of the probability of hazard (see the discussion in Section IIA) and thus of expenditures on self-protection, our analysis of market insurance implies that self-insurance is likely to create a moral hazard. That is to say, the availability of self-insurance would discourage self-protection and vice versa. Moreover, technological progress in the provision of one would tend to discourage the other.

This analysis of moral hazard applies not only to the relation between self-protection and insurance as ordinarily conceived, but also to the relation between protection and insurance for all uncertain events that can be influenced by human actions. For example, do unemployment compensation, relief, or negative income tax rates increase the probability that someone becomes unemployed? Does the presence of underground shelters increase the probability that a country goes to war, the use of seat belts the probability of an automobile accident, or generous parental support the probability that children become "irresponsible"?

Since each of these, in effect, relates a form of insurance to a form of protection, our answers are not necessarily "yes," and depend on how responsive the cost of insurance is to the amount spent on protection. Shelters and seat belts are ways to self-insure, and have costs that are essentially unrelated to the probability of the hazards; therefore they would tend to reduce (perhaps only slightly) the incentive to avoid a war or an automobile accident. On the other hand, if the cost—in time, embarrassment, etc.—of applying for relief, unemployment compensation, or parental support were sufficiently positively related to its frequency, the answers might well be "no": the availability of insurance might encourage the insured to make his own efforts.

Appendix A

Self-Insurance and Market Insurance

If both self-insurance and market insurance are available, the expected utility is

$$U^* = (1 - p)\, U(I_1{}^e - c - s\,\pi) + p\, U(I_1{}^e - L(L^e, c) - c + s). \quad \text{(A1)}$$

The values of c and s that maximize this function must satisfy the first-order optimality conditions

$$U^*{}_s = -(1 - p)\, U_1{}'\,\pi + p\, U_0{}' = 0 \quad \text{(A2)}$$

$$U^*{}_c = -(1 - p)\, U_1{}' - p\, U_0{}'[L'(c^*) + 1] = 0. \quad \text{(A3)}$$

Clearly, equation (A3) would be satisfied only if $\delta = [L'(c^*) + 1] < 0$: only if expenditures on self-insurance increased the net income in the hazardous state.

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Second-order optimality conditions are

$$U^*_{ss} = (1 - p)\, U_1'' \pi^2 + p\, U_0'' < 0 \tag{A4}$$

$$U^*_{cc} = (1 - p)\, U_1'' + p\, U_0'' \delta^2 - p\, U_0' L'' < 0 \tag{A5}$$

$$\Delta = U^*_{ss}\, U^*_{cc} - (U^*_{sc})^2 > 0. \tag{A6}$$

Equations (A4) and A5) are obviously satisfied if everywhere $U'' < 0$ and $L'' = \partial^2 L/\partial c^2 > 0$, that is, if the marginal utility of income and the marginal productivity of self-insurance are both decreasing. These assumptions are also sufficient to satisfy equation (A6) since

$$U^*_{sc} = (1 - p)\, U_1'' \pi - p\, U_0'' \delta < 0. \tag{A7}$$

Utilizing the first-order condition $\pi \delta = -1$, we can write

$$\Delta = -p(1 - p)\, U_0'\, U_1''\, L''\, \pi^2 - p^2\, U_0'\, U_0''\, L'',$$

which is positive if $U'' < 0$ and $L'' > 0$.

A. *Terms of Trade Effects*

The effect of an increase in π on the optimal values of s and c—$I_1{}^e$, L^e, and p held constant—can be found by differentiating equations (A2) and (A3) with respect to π. By Cramer's rule,

$$\frac{ds^*}{d\pi} = \frac{A_1 U^*_{cc} - A_2 U^*_{sc}}{\Delta} = \frac{1}{\Delta}[(1 - p)^2 U_1' U_1'' + p(1 - p) U'_1 U_0'' \delta^2 - p(1 - p) U_0' U_1' L'' + p(1 - p) U_0' U_1'' s^* \pi L''] = \frac{(-)}{(+)} < 0 \tag{A8}$$

where $-A_1 = -(1 - p)U_1' + (1 - p)U_1''\, s^*\, \pi$ and $-A_2 = (1 - p)U_1''\, s^*$ are the partial derivatives of (A2) and (A3) with respect to π. Similarly,

$$\frac{dc^*}{d\pi} = \frac{A_2 U^*_{ss} - A_1 U^*_{sc}}{\Delta} = \frac{1}{\Delta}[-(1 - p)^2 U_1' U_1'' \pi + p(1 - p) U_1' U_0'' \delta] = \frac{(+)}{(+)} > 0. \tag{A9}$$

Hence, market insurance and self-insurance can be considered substitutes.

By similar reasoning, the effect of an increase in p on the optimal values of s and c, given that λ in $\pi = [(1 + \lambda)p]/(1 - p)$, $I_1{}^e$ and $I_1{}^e$ are constant, is found to be

$$\frac{ds^*}{dp} = \frac{B_1 U^*_{cc} - B_2 U^*_{sc}}{\Delta} = \frac{1}{\Delta}[U_0' U_1'' L'' s^* \pi^2 + (1 - p) U_1' U_1'' \pi + (1 - p) U_0' U_1'' - p U_1' U_0'' \delta + p U_0' U_0'' \delta^2] = \frac{(-)}{(+)} < 0; \tag{A10}$$

also

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$$\frac{dc^*}{dp} = \frac{B_2U^*_{cc} - B_1U^*_{sc}}{\Delta} = \frac{1}{\Delta}\,[-(1-p)U_1'U_1''\pi^2 - (1-p)U_0'U_1''\pi - pU_1'U_0'' + pU_0'U_0''\delta] = \frac{(+)}{(+)} > 0 \tag{A11}$$

where $-B_1 = (1-p)U_1''s^*\pi'(p)\pi$, $-B_2 = U_1' - U_0'\delta + (1-p)U_1''s^*\pi'(p)$, and $\pi'(p) = \partial\pi/\partial p = \pi/[p(1-p)]$.

B. An Endowment Effect

The effect of a decrease in $I_0^e = I_1^e - L^e$—I_1^e, π, p, and L' held constant—can be shown to be

$$-\frac{ds^*}{dI_0^e} = \frac{-p^2U_0'U_0''L''}{\Delta}\,\frac{\partial L}{\partial L^e} = \frac{(+)}{(+)} > 0, \tag{A12}$$

where, by assumption, $\partial L/\partial L^e > 0$; and

$$-\frac{dc^*}{dI_0^e} = \frac{-pU_0''\delta U^*_{ss} - pU_0''U^*_{sc}}{\Delta}\,\frac{\partial L}{\partial L^e} = 0, \tag{A13}$$

since by equations (A7) and (A8) $U^*_{sc} = -\delta U^*_{ss}$. If the change in I_0^e also changed L', the results would be different.

Appendix B

Self-Protection and Market Insurance

If both market insurance and self-protection are available, the expected utility is

$$U^* = [1 - p(p^e, r)]\,U(I_1^e - r - s\,\pi(r)) + p(p^e, r)U(I_0^e - r + s). \tag{B1}$$

The first-order optimality conditions are

$$U^*_s = -(1-p)U_1'\pi + pU_0' = 0 \tag{B2}$$

$$U^*_r = -p'(r^*)(U_1 - U_0) - (1-p)U_1'[1 + s^*\pi'(r)] - pU_0' = 0 \tag{B3}$$

where $p'(r^*) < 0$ and $\pi'(r^*) \leqslant 0$.

Second-order conditions are that

$$U^*_{ss} = (1-p)U_1''\pi^2 + pU_0'' < 0 \tag{B4}$$

$$U^*_{rr} = -p''(r^*)(U_1 - U_0) + (1-p)U_1''[1 + s^*\pi'(r^*)^2 + pU_0'' \tag{B5}$$

$$+ 2p'(r^*)\{U_1'[1 + s^*\pi'(r)] - U_0'\} - (1-p)U_1's^*\pi''(r^*) < 0. \tag{B6}$$

$$\Sigma = U^*_{ss}U^*_{rr} - (U^*_{sr})^2 > 0.$$

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Equations (B4) and (B5) would be satisfied if $U'' < 0$, if both $p''(r^*)$ and $\pi''(r^*) > 0$, and if $D = 2p'(r^*)\{U_1'[1 + s^*\pi'(r^*)] - U_0'\}$ (which is positive if $U'' < 0$)[47] were small in absolute value relative to the other terms in equation (B5). These conditions are also sufficient to satisfy equation (B6) if, in particular, $D < (1 - p)U_1' s^* \pi''(r^*) + p''(r^*)(U_1 - U_0)$.

Since $\pi = [(1 + \lambda)p]/(1 - p)$, the effect of an increase in r on π would be

$$\pi'(r) = \frac{(1 + \lambda)p'(r)}{(1 - p)^2} + \frac{p\lambda'(r)}{(1 - p)}, \tag{B7}$$

where $\lambda'(r)$ gives the effect on the loading of an additional expenditure on self-protection and is generally assumed to be positive.[48]

If insurance were always available at an actuarially fair price, then $\lambda(p) = 0$ for all p; hence

$$\pi = \frac{p(r)}{1 - p(r)}, \quad \pi'(r) = \frac{p'(r)}{(1 - p)^2}, \quad \text{and}$$

$$\pi''(r) = \frac{p''(r)}{(1 - p)^2} + \frac{2[p'(r)]^2}{(1 - p)^3} > 0.$$

Equation (B2) reduces to

$$U_1' = U_0', \tag{B8}$$

and equation (B3) to

$$p'(r^*) = -\frac{(1 - p)}{s^*}. \tag{B9}$$

Therefore, $\pi'(r^*) = -1/[s^*(1 - p)]$ and $1 + s^*\pi'(r^*) = -\pi$.

A. *Terms of Trade Effects*

If an initially fair price $\pi = [p(r^*)]/[1 - p(r^*)]$ were increased by an increase in the loading with no change in $\pi'(r)$,[49] the change in the optimal values of s and r would be given by

$$\frac{ds^*}{d\pi} = \frac{C_1 U^*_{rr} - C_2 U^*_{sr}}{\Sigma} = \frac{C_1\left[\dfrac{2U_1'}{s^*} - (1 - p)U_1' s^* \pi''(r)\right]}{\Sigma}$$

$$= \frac{(-)}{(+)} < 0,[50] \tag{B10}$$

[47] According to equations (B2) and (B9) and the condition $U'' < 0$, $U_1'[1 + s^*\pi'(r^*)] < U_0'$ if $\pi \gtrless p/1 - p)$.

[48] One can write $\lambda'(r) = (\partial\lambda/\partial p)(\partial p/\partial r$, where $\partial p/\partial r < 0$. Hence, $\lambda'(r) > 0$ only if $\partial\lambda/\partial p < 0$. (But see our discussion in Section IC.)

[49] According to equation (B7), an increase in λ due to an exogenous factor θ would not change $\pi'(r)$ if, and only if, $[p'(r)/(1 - p)][\partial\lambda(r, \theta)/\partial\theta] = -p[\partial\lambda'(r, \theta)/\partial\theta]$. This assumption is made to separate an autonomous change in the price of insurance from an autonomous change in the effect of self protection on the price of insurance.

[50] Using equations (B8) and (B9) and the second-order optimality conditions

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$$\frac{dr^*}{d\pi} = \frac{C_2 U^*_{ss} - C_1 U^*_{sr}}{\Sigma} = \frac{-2(1-p)U_1' U^*_{sr}}{\Sigma} = \frac{(-)}{(+)} < 0, \tag{B11}$$

where

$$-C_1 = U^*_{s\pi}\Big]_{\substack{s^*,\, r^*,\, p,\, \pi'(r^*) \\ \text{constant}}} = -(1-p)U_1' + (1-p)U_1'' s^* \pi$$

and

$$-C_2 = U^*_{r\pi}\Big]_{\substack{s^*,\, r^*,\, p,\, \pi'(r^*) \\ \text{constant}}}$$

$$= p'(r^*)U_1' s^* + (1-p)U_1'' s^* [1 + s^* \pi'(r^*)]$$

$$= -(1-p)U_1' - (1-p)U_1'' s^* \pi$$

(from equation [B9]). Hence, if the price of insurance increased from an initially fair level, the demand for both self-protection and market insurance would decrease.

If the price of insurance were always actuarially fair, the effect of an exogenous increase in the productivity of self-protection on s^* and r^* with no change in the endowed probabilities and in the endowed incomes would be given by

$$\frac{ds^*}{d\beta} = \frac{D_2 U^*_{sr}}{\Sigma} = \frac{(+)}{(+)} > 0 \tag{B12}$$

and

$$\frac{dc^*}{d\beta} = \frac{-D_2 U^*_{ss}}{\Sigma} = \frac{(+)}{(+)} > 0, \tag{B13}$$

where $D_2 = -U_1'[s/(1-p)]\partial p'/\partial\beta$ and, by assumption $\partial p'/\partial\beta < 0$. Technological improvements in self-protection are thus seen to increase the demand for both market insurance and self-protection.

B. *The Effect of Exogenous Changes in* p *and* L

If insurance were provided at an actuarially fair price, and if the endowed probability increased due to an exogenous factor γ with no change in $p'(r)$, then

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discussed above, it follows that $U^*_{rr} = (1-p)\ U_1''\pi^2 + pU_0'' + 2U_1'/s^* - (1-p)\ U_1' s^* \pi''\ (r) < 0$; $U^*_{sr} = -(1-p)\ U_1''\pi^2 - pU_0'' > 0$; and

$$\Sigma = U^*_{ss}\left[\frac{2U_1'}{s^*} - (1-p)U_1'\, s^*\, \pi''(r)\right] > 0.$$

Since by equation (B4) $U^*_{ss} < 0$, $2U_1'/s^* - (1-p)\ U_1' s^* \pi''\ (r)$ must be negative in order for Σ to be positive.

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$$\frac{ds^*}{d\gamma} = \frac{\left[-(1-p)U_1''s^*\pi\,\pi'(p)\right]\left[\dfrac{2U_1'}{s} - (1-p)U_1's^*\pi''(r)\right]}{\Sigma}$$

$$\frac{\partial p}{\partial \gamma} = \frac{(-)}{(+)} < 0 \tag{B14}$$

where, by assumption, $\partial p/\partial\gamma > 0$, and

$$\frac{dr^*}{d\gamma} = \frac{(1-p)U_1''s^*\pi\,\pi'(p)[U^*_{ss} + U^*_{sr}]}{\Sigma}\frac{\partial p}{d\gamma} = 0, \tag{B15}$$

since $U^*_{ss} = -U^*_{sr}$.[51] The last result is intuitively obvious since a fair price of insurance implies that $-p'(r^*) = 1/(I_1^e - I_0^e)$; therefore, r^* is independent of p provided that $p'(r)$ is unaffected by changes in p^γ. By the same reasoning one can show that an increase in the size of the prospective loss increases the optimal values of both s and r.

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[51] See equation (B4) and the footnote following equation (B10).

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